

SBN Explorer

Methodology

We consider the complete constrained architecture space $\Omega = \mathbb{B}^n$, where each configuration $\mathbf{c} = (C_1, \dots, C_n) \in \Omega$ is assigned a scalar fitness value through a function $f : \mathbb{B}^n \rightarrow \mathbb{R}$. The methodology consists of six stages, from local effect analysis to global variance attribution.

1 Stage 0: Multi-Level Discretization

When categorical analyses are required, the scalar fitness is discretized into performance classes:

- **Binary performance** (Fit_bin): split at the median fitness value;
- **Elite performance** (Fit_elite): restriction to the top 10–20% of SBNs.

These two discretizations respectively capture coarse structural contrasts and highly selective architectural regimes.

2 Stage 1: Hypercube Edge Analysis

For each constraint C_i , we evaluate its marginal effect over all 2^{n-1} contexts induced by the remaining constraints. Writing $C_{-i} \in \mathbb{B}^{n-1}$, define

$$\Delta f_i(C_{-i}) = f(C_{-i}, C_i = 1) - f(C_{-i}, C_i = 0).$$

The resulting family $\{\Delta f_i(C_{-i})\}_{C_{-i} \in \{0,1\}^{n-1}}$ is summarized by:

- the mean effect $\bar{\Delta}_i = 2^{-(n-1)} \sum_{C_{-i}} \Delta f_i(C_{-i})$;
- the effect variance $\sigma_{\Delta_i}^2 = 2^{-(n-1)} \sum_{C_{-i}} (\Delta f_i(C_{-i}) - \bar{\Delta}_i)^2$;
- the sign stability, i.e., the proportion of contexts for which $\Delta f_i(C_{-i}) > 0$.

Large $\sigma_{\Delta_i}^2$ relative to $|\bar{\Delta}_i|$ indicates context dependence.

3 Stage 2: Epistasis Matrix

For each pair (C_i, C_j) and each context $C_{-\{i,j\}} \in \mathbb{B}^{n-2}$, define

$$\varepsilon_{ij}(C_{-\{i,j\}}) = f_{11} - f_{10} - f_{01} + f_{00},$$

where $f_{ab} = f(C_i = a, C_j = b, C_{-\{i,j\}})$.

The pairwise interaction is summarized by

$$\begin{aligned}\bar{\varepsilon}_{ij} &= 2^{-(n-2)} \sum_{C_{-\{i,j\}}} \varepsilon_{ij}(C_{-\{i,j\}}), \\ \sigma_{\varepsilon_{ij}}^2 &= 2^{-(n-2)} \sum_{C_{-\{i,j\}}} (\varepsilon_{ij}(C_{-\{i,j\}}) - \bar{\varepsilon}_{ij})^2.\end{aligned}$$

This yields the epistasis matrix $\mathbf{E} = [\bar{\varepsilon}_{ij}]_{i,j=1}^n$.

Stable positive values identify synergistic pairs; stable negative values identify antagonistic pairs; large variances indicate context-dependent interactions.

4 Stage 3: Formal Concept Analysis (FCA)

To extract minimal structural patterns associated with high fitness, we define, for each discretization of Stage 0, a formal context $\mathbb{K} = (G, M, I)$, with

- $G = \Omega = \mathbb{B}^n$;
- $M = \mathcal{C} \cup \{\text{high-performance}\}$, where $\mathcal{C} = \{C_1, \dots, C_n\}$;
- $I \subseteq G \times M$ the incidence relation.

A formal concept is a pair (A, B) such that

$$\begin{aligned}A' &= \{m \in M \mid \forall g \in A, (g, m) \in I\} = B, \\ B' &= \{g \in G \mid \forall m \in B, (g, m) \in I\} = A.\end{aligned}$$

The concept lattice $\mathfrak{B}(\mathbb{K})$ is ordered by

$$(A_1, B_1) \leq (A_2, B_2) \iff A_1 \subseteq A_2 \quad (\text{equivalently } B_2 \subseteq B_1).$$

From each lattice, we extract:

- minimal generators: minimal subsets $B \subseteq \mathcal{C}$ such that $B \cup \{\text{high-performance}\}$ is closed;
- association rules $B_1 \rightarrow B_2$ with support

$$\text{sup}(B_1 \cup B_2) = \frac{|B'_1 \cap B'_2|}{|G|}$$

and confidence

$$\text{conf}(B_1 \rightarrow B_2) = \frac{|B'_1 \cap B'_2|}{|B'_1|}.$$

These objects provide minimal structural signatures of high-performance regions.

5 Stage 4: Post-Lattice Validation

Each pattern $\mathcal{M} \subseteq \mathcal{C}$ extracted at Stage 3 is tested on the restricted subcube

$$(\mathbb{B}^n)^{\mathcal{M}} = \{C \in \mathbb{B}^n \mid \forall C_i \in \mathcal{M}, C_i = 1\}.$$

For each $C_j \notin \mathcal{M}$, we evaluate whether adding C_j significantly improves fitness by applying a Wilcoxon signed-rank test to the paired differences $\Delta f_j(C_{-j})$ within the restricted subspace. If no significant improvement is detected ($p > 0.05$), then C_j is treated as locally neutral relative to $\mathcal{M}$.

This step distinguishes structurally sufficient patterns from incomplete ones.

6 Stage 5: Spectral Variance Decomposition on the Hypercube

Since f is observed on the full hypercube, all interaction orders are exactly identifiable. We therefore use an orthogonal Walsh–Hadamard decomposition.

6.1 Stage 5a: Walsh–Hadamard Decomposition

We re-encode each binary variable $C_i \in \{0, 1\}$ into centered coordinates: $x_i = 2C_i - 1 \in \{-1, +1\}$. The fitness function admits a unique expansion over the Walsh basis:

$$f(x) = \sum_{S \subseteq \{1, \dots, n\}} \hat{f}(S) \chi_S(x), \quad \chi_S(x) = \prod_{i \in S} x_i,$$

with coefficients $\hat{f}(S) = 2^{-n} \sum_{x \in \{-1, 1\}^n} f(x) \chi_S(x)$.

Orthogonality gives $\sum_{x \in \{-1, 1\}^n} \chi_S(x) \chi_T(x) = 2^n \delta_{ST}$, hence

$$\text{Var}(f) = \sum_{S \neq \emptyset} \hat{f}(S)^2.$$

For each order k , define the contribution of interaction order k and the normalized epistatic spectrum, respectively

$$V_k = \sum_{|S|=k} \hat{f}(S)^2, \quad \eta_k^2 = \frac{V_k}{\sum_{m=1}^n V_m}.$$

Thus:

- $k = 1$ corresponds to additive effects;
- $k = 2$ to pairwise epistasis;
- $k \geq 3$ to higher-order interactions.

6.2 Stage 5b: Restricted Hierarchical Model

For interpretability, we introduce a truncated descriptive model retaining:

- [1] all first-order terms;
- [2] the dominant second-order terms;
- [3] indicator terms associated with selected FCA patterns $\mathcal{M}_k$.

This yields

$$\tilde{f}(x) = \mu + \sum_{|S|=1} \hat{f}(S) \chi_S(x) + \sum_{S \in \mathcal{S}_2} \hat{f}(S) \chi_S(x) + \sum_k \alpha_k \mathbb{I}[x \in \mathcal{M}_k], \quad (1)$$

where $\mathcal{S}_2$ is the set of dominant pairwise terms.

This model is descriptive only. Exact variance attribution remains given by the Walsh spectrum.

6.3 Final Artifact: Contribution Spectrum

For each nonempty component S , define

$$\eta_S^2 = \frac{\hat{f}(S)^2}{\sum_{T \neq \emptyset} \hat{f}(T)^2}.$$

This yields a complete contribution spectrum comprising:

- additive effects ($|S| = 1$),
- pairwise interactions ($|S| = 2$),
- higher-order couplings ($|S| \geq 3$).

FCA-derived patterns are reported separately through their support, confidence, and conditional fitness gains.

7 Methodological Synthesis

The methodology proceeds from local to global structure:

- Stage 1: marginal effects;
- Stage 2: pairwise epistasis;
- Stage 3: minimal structural rules;
- Stage 4: local validation of these rules;
- Stage 5: exact variance decomposition by interaction order.

It therefore connects exhaustive combinatorial exploration, lattice-based structural abstraction, and exact spectral attribution within a single framework. The methodology progresses from *combinatorial exhaustivity* (Stages 0–2) through *structural abstraction* (Stages 3–4) to *quantitative attribution* (Stage 5).